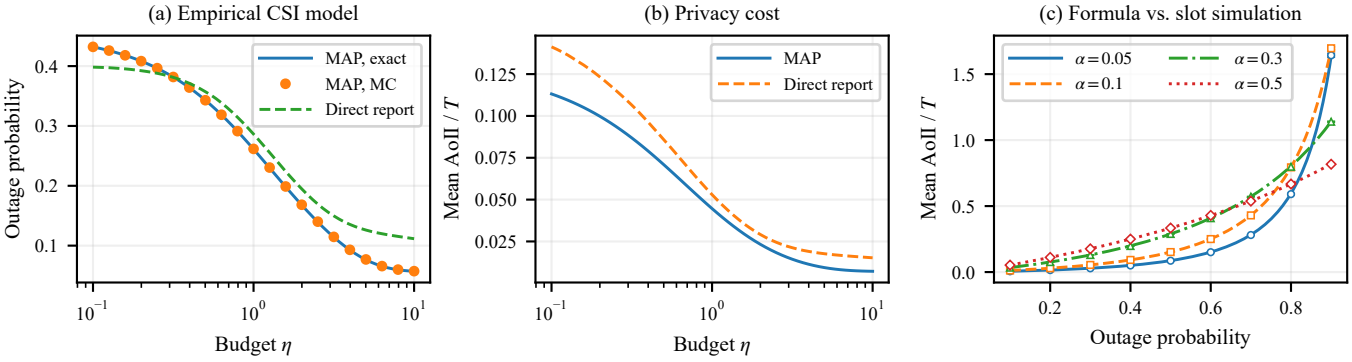
(a) Empirical CSI model
Outage probability
0.4
0.3
0.2
0.1
10^{-1}
10^{0}
10^{1}
Budget η
MAP, exact
MAP, MC
Direct report
(b) Privacy cost
Mean AoII / T
0.125
0.100
0.075
0.050
0.025
10^{-1}
10^{0}
10^{1}
Budget η
MAP
Direct report
(c) Formula vs. slot simulation
Mean AoII / T
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
Outage probability
$\alpha = 0.05$
$\alpha = 0.1$
$\alpha = 0.3$
$\alpha = 0.5$